Introduction to Artificial Intelligence

1. Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing images, making decisions, solving problems, and learning from data.

2. Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence. Instead of explicitly programming every rule, machine learning algorithms learn patterns from data and use those patterns to make predictions or decisions.

3. Deep Learning

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers. Deep learning is widely used in image recognition, speech recognition, natural language processing, and generative AI.

4. Natural Language Processing

Natural Language Processing (NLP) is a branch of AI that focuses on enabling computers to understand, process, analyze, and generate human language. Applications include sentiment analysis, translation, chatbots, text classification, and question answering.

5. Generative AI

Generative AI refers to systems that can create new content such as text, images, audio, video, and code. Large Language Models (LLMs) are an important example of generative AI systems designed to understand and generate natural language.

6. Large Language Models

A Large Language Model (LLM) is a neural network trained on large amounts of text data. It learns statistical and semantic patterns in language and can generate text, answer questions, summarize information, translate languages, and assist with programming.

7. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with a language model. Relevant information is retrieved from an external knowledge source and provided to the language model as context before generating an answer. This helps the system answer questions using information that may not have been present in the model's original training data.

8. Artificial Intelligence Course

An Artificial Intelligence course may include Python, Machine Learning, Deep Learning, NLP, and Generative AI. A structured course can progress from programming fundamentals to machine learning, neural networks, transformers, LLMs, embeddings, vector databases, and RAG applications.

9. Course Duration

The Artificial Intelligence course described in this document has a duration of 6 months.